# M.Tech (Integrated) Computer Science and Engineering
# CSI4901–Capstone Project

## HOSPITAL MANAGEMENT SYSTEM USING MERN STACK

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# ABSTRACT


In recent times, the software applications provide the services faster than traditional services. The appointment in the hospital takes a long time and you have to wait for your turn to visit the doctor. It is such a long and time taking process. Instead, we can do the same thing through the latest software technologies to maintain appointments, prescriptions and medicines. The main of this project is to provide efficient approach to reduce waiting time, easy schedules, payments, and doctor visits. By leveraging MERN Stack technologies, which provides both front end and back end services, we intend to create web applications for both doctors and patients for scheduling appointments.

Our solution gives the three main actors - Admin, Doctor, and Patient, a user-friendly user interface (UI) with the goal of enhancing the medical industry. The technology simplifies numerous administrative and medical procedures by combining different functions into a single platform, guaranteeing improved patient care and effective hospital workflows. The Admin role is crucial in maintaining the system's operational backbone. Admins can manage doctor schedules by allotting available timings and ensuring that patients can easily book appointments based on the availability of healthcare professionals. In addition, admins oversee the management of medicine inventories, ensuring that stock levels are adequate and medicines are available for dispensing when prescribed by doctors.

The next module lets doctors schedule visits, give prescriptions, and provide treatments is available to them. Doctors may effectively plan their workday by using the system to view their schedules in real time. Doctors can digitally write prescriptions while seeing patients, and the prescriptions are automatically stored in the system for later use. Additionally, they oversee the distribution of medications, guaranteeing that patients obtain the dosages that are advised. Prescriptions and medical information are integrated, which increases accuracy and decreases paperwork while improving patient care.

Patients can easily communicate with the hospital through to the Patient module. Based on the admin-allotted schedules, patients can log into the system to schedule appointments with clinicians who are available. By doing away with the necessity for manual appointment scheduling, this self-service tool simplifies and speeds up the process. In order to improve accessibility, patients can also check their prescription history, assuring that their medical records are accessible at all times. HMS is a MERN Stack application that contains Mongo DB, Express JS, React JS, and Node JS to build a full stack web application for patients and doctors.

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# 1. INTRODUCTION

A Hospital Management System (HMS) is a comprehensive software platform designed to manage the operations of a hospital or healthcare facility. It incorporates a number of healthcare administration functions, such as prescription management, medical inventory management, doctor scheduling, and patient visits. The effectiveness, precision, and caliber of healthcare services are enhanced by an HMS through the digitization and automation of clinical and administrative processes. The MERN stack, which consists of Node.js, Express.js, React, and MongoDB, is used to build this HMS. Because of the MERN stack's efficiency, scalability, and flexibility, full-stack web applications are developed using it extensively. Every element in the stack is essential to creating a cutting-edge, adaptable, and data-driven hospital management system:

- **MongoDB** serves as the NoSQL database to store and manage structured and unstructured healthcare data.
- **Express.js** provides the server-side framework to handle API requests and manage business logic.
- **React.js** is used for building the user-friendly frontend, ensuring a seamless user experience for admins, doctors, and patients.
- **Node.js** handles the backend operations, ensuring fast and efficient communication between the frontend and database.

## 1.1 BACKGROUND

The implementation of technology in the healthcare sector to improve patient care and streamline operations has grown significantly. Manual procedures were a major part of traditional hospital administration, such as phone-based appointment scheduling, paper-based records, and manual inventory tracking. These approaches frequently resulted in inefficiencies including lost paperwork, incorrect appointment scheduling, and trouble keeping up with the massive amount of medical data. It became essential to upgrade these procedures by implementing digital technologies as healthcare services grew.

To overcome these obstacles, a Hospital Management System (HMS) was created. Key hospital functions like prescription handling, inventory management, medical record administration, physician scheduling, and patient registration are all automated by HMS software. As web-based applications have grown in popularity, HMS platforms are now reachable from any location and offer real-time data to patients, physicians, and hospital management. This change has enhanced overall patient care, decreased human error, and increased operational efficiency.

## 1.2 MOTIVATIONS

The key motivations of the projects include:

- **To increase operational efficiency:** Hospital employees can concentrate on patient care by reducing administrative duties by automating processes like inventory management, prescription handling, and appointment scheduling.
- **To improve patient experience:** Convenience and happiness are increased when patients have simple access to appointments, medicines, and medical records via an intuitive interface.
- **To have real-time data access:** Improving coordination and accelerating decision-making is ensured by providing doctors, patients, and administrators with real-time access to vital data.
- **To minimize human error:** Digitizing documents and procedures lowers the possibility of errors, increasing precision in domains like patient data and inventory control.
- **To ensure flexibility and scalability:** The MERN stack makes it possible for the system to be readily expanded to meet the growing demands of hospitals, enabling it to adapt to changing healthcare contexts.

## 1.3 SCOPE OF THE PROJECT

- **Appointment Management:** Patients can book, view, and cancel appointments with available doctors. Admins can manage doctor schedules and view overall appointment reports. Doctors can view their schedules and update appointment statuses.
- **Prescription and Inventory Management:** Prescriptions for patients are created, updated, and viewed by doctors. Patients have access to their medical information and prescription histories. The system ensures real-time stock updates to avoid shortages.
- **User Authentication:** Secure login and role-based access control for Admins, Doctors, and Patients. Admins can manage user accounts and assign roles.

# 2. PROJECT DESCRIPTION AND GOALS

## 2.1 LITERATURE REVIEW

**[1]** Smart software applications can manage medical patient appointments in a very effective way. With the use of smart technologies, UAE developed a website that contains mobile app, website and their own database to store the patient's information. . It integrates technologies like C#, MySql to complete the website to provide appointments to the patients. The proposed methodology Embedded Expressed Shell(EES) and AI to solve the diagnostic problems using human intelligence and also the doctor's advice to solve the patient's health problems. EES was integrated into the application by using Visual Studio with C# to create the inference engine. This proposed system has been verified using different test cases to check how it can improve further. User logins to website and sends a query and the system answers to the query according to the data available. Overall, the website is successfully giving 75% current suggestions to the users and result of the process is called as Inference Engine (IE).

**[2]** The Mobile Medical System is designed especially for families to take care of their health in an organized way. The application is designed to store records of patients such as appointments and prescriptions given by doctor. The proposed methodology uses MERN Stack technologies where MERN represents MongoDB, Express JS, React JS, and Node JS. MERN is a full stack application which has both front end and back end. The application is user friendly, which keeps track of remainders, appointment and prescriptions. Along with MERN Stack, Raspberry Pi 4 is used for the sake of deployment. It is a minicomputer to build our application and also helps to access the information. Overall, the application is user friendly, scalable and robust. The main goal of the paper is to create a application to store prescriptions of the patients so that user can have clear overview of their health.

**[3]** This paper discusses the development of websites and web applications to manage the booking medical appointments. The proposed methodology is SCRUM and Django framework. This framework allows the developers to program in python language and has MVC. SCRUM is an iterative development process model. The work of the SCRUM process is divided into 4 subparts and each team has to do the allotted task in certain period of time. MVC handles business logic, user interaction and shows the data to the user. The application has CRUD operations in the admin side login. Overall, the final website is user friendly, gives fast response and manages the integrity of the application.

**[4]** This paper discusses the improvement of medical system and implementation of an expert system for hospital appointment scheduling. It employs the technologies of internet based expert systems such as MYCIN, PUFF, CASNET, etc to deliver the application early. These expert systems improve the doctor patient interactions, and reduce the waiting time, and saves time for both doctor and patient. The patients can book an appointment with the help of internet and the ones who don't the exact department to choose are directed to embedded systems in the application. The decision trees are made to categorize the patient into correct department. SQL and ASP.NET is used for application development and queries. Overall, the internet based applications has potential advantages in healthcare.

**[5]** This literature paper gives an overview of accurate approach to improve the vaccination appointment system in vaccination centers. It implements the integration of database technologies with mobile computing to improve the quality of service for patients, medical staff and doctors. The paper gives the overview of how to manage vaccination process for children and to keep track of the vaccination and other medical records. The study also proposed an Android-based mobile application and a web application to book an appointment and to maintain and manage health records of the patients. The application is designed to give importance for both patients and medical staff, and its goal is to reduce the patient waiting time and improve healthcare. The paper also gives the overview of the proposed architecture, where the architecture highlights different layers used such as Resource, Service and Presentation layers. Also, they mentioned related works in healthcare system and it's used and is grateful to the Palestine Technical University for publishing their research.

**[6]** This paper mentions the disadvantages of booking medical appointments for nuclear medicine examinations in a large Chinese hospital. The paper addresses the problems related to radiopharmaceuticals and the behavior of the patients. The authors of the paper propose a methodology to schedule online appointments using the following to reduce the disadvantages. The schedules in the application are categorized as FCFS, Offopt, Offline model and dynamic model. Dynamic is considered due to threshold level. They also conducted various test to show the effectiveness of the proposed methodology, showing the major improvements in result examination, resource utilization, and timely rate. The paper also explains how other algorithms like genetic algorithms and the calculation of sensitivity analysis can improve the booking of appointments in nuclear examinations.

**[7]** This paper explains the disadvantages of scheduling the patient appointments within a moving booking window. To address this, they introduced a Markov Decision Process(MDP) model to maximize the reward and also reduces the capacity cost. Along MDP, they also used FCFS, STP, MP, and ASP for comparison purposes. The best practices are defined, as are structural aspects of the optimal advance scheduling strategy, such as the multimodularity and supermodularity. It proposes efficient policies based on the results from conducted tests to compare the efficiency and effectiveness of the MDP. As for rewards, multiple booking thresholds are preferred. The two policies assignment sequence policy and multiple threshold policy are proposed for simulation. This paper also examines the advantages of the proposed methodology to improve the schedules in moving booking window.

**[8]** This paper gives an overview of the impact of creating an medical booking system for outpatient clinics. The designed application was user friendly and secure. The proposed methodology includes various technologies such as PHP, MySql. To check the status of the application, the evaluation metrics compares the results before and after the implementation in ten clinics. The evaluation metrics used are patient waiting time, doctor available time, service time, no show rate, etc. The analysis made 95% of confidence level which concludes that there is less waiting time for patients and increased doctor availability in the clinic. Hence, the online appointment scheduling algorithm was successful in implementing various performance metrics and improved medical care for patients. Due to the successful implementation, they created app for both iOS and android versions.

**[9]** This literature paper identifies the challenges of scheduling medical appointments and proposes a model that increases efficiency in booking the appointments to decrease no show which interrupts the healthcare system. This paper uses machine learning models to calculate the patient attendance. Decision Tree, Neural Network, Support Vector Machine, Linear Regression, Naive Bayes are used to perform predictions. The success rate of proposed models are calculated by various performance metrics such as accuracy, precision, recall, F1 score. The application was created using AMPL language and solved using CPLEX solver. Overall, this paper discusses the use of machine learning algorithms to predict patient attendance and improves the challenge of appointment scheduling by creating time slots with overbooking strategy.

**[10]** This literature paper mainly focuses on optimizing the medical appointment scheduling for outpatients in a diagnostic centers, particularly for (Single Photon Emission Computed Tomography (SPECT) scanning purposes. It explains the radiopharmaceutical compound that can be delivered to the clinics in a very short period of time which can minimize violations and maximize the resource availability. The study introduces a dynamic robust optimization framework to handle these challenges and validates the model using data from a molecular imaging center in Tehran, Iran. It reviews important studies related to radiopharmaceutical administration and appointment scheduling in medical imaging centers. The paper also addresses the details of molecular imaging procedure during the operation, stating the connection of components involved in the imaging process.

## 2.2 GAPS IDENTIFIED

- Not all health organizations are connected, which means users might still have to log in to different websites for certain healthcare services [2].
- The system may not be accessible to individuals with limited internet access or technological proficiency [4].
- The system's reliability in accurately directing patients to the right department based on their symptoms might be a challenge, especially if the symptoms are complex or require in-person evaluation [5].
- The responsive of the application needs to be implemented to boost the application performance and to increase the availability of the website [5].
- Scalability to real world hospital scenario where large number of patients who want to access the application and requires complex scheduling requirements [7].

## 2.3 OBJECTIVES

- To integrate the MERN stack to enable scalability and flexibility, making it simple to integrate new features and expand the system in the future.
- To increase efficiency in administration by providing resources for tracking inventory, scheduling appointments, checking doctor availability, and creating hospital reports.
- To improve patient care by providing a simple user friendly interface to the patients to schedule appointments, read prescriptions, and access medical data.
- To enable real-time data access for doctors, admins, and patients, ensuring fast access to the website.

## 2.4 PROBLEM STATEMENT

Ineffective manual processes that hospitals frequently use to manage appointments, medications, and inventory can cause mistakes, increase in waiting time, and operational obstacles. Doctors lack simplified resources to manage schedules and medicines, while patients struggle to make appointments and access their medical records. Hospital administrators are burdened with manual tracking of inventory and doctor availability, resulting in poor resource management. These issues negatively impact the quality of patient care and operational efficiency. Therefore, there is a need for a centralized, automated, and user-friendly system to address these challenges. Therefore, there is a need for a comprehensive, web-based Hospital Management System (HMS) that uses latest software technologies such as MERN Stack, which help to automate key hospital functions, improves operational efficiency, and enhances patient care while ensuring data security and scalability.

## 2.5 PROJECT PLAN

**Phase 1: Project planning and brainstorming (July)**

- **July 15-22:** Find a problem domain and start gathering the requirements to initialize the project.
- **July 23-31:** Start creating a Github repository and plan the timeline of the project.

**Phase 2: Review 0 and Literature Survey (August)**

- **Week 1:** Review 0 submission to the guide.
- **Week 2-4:** Collecting quality research papers for literature survey by cross checking the papers with dblp (https://dblp.org/) website and pushing them to Github.

**Phase 3: Finding gaps in literature survey (September)**

- **Week 1-2:** Collect the gaps in the literature review and form a problem statement.
- **Week 3-4:** Frame objectives and scope of the project.

**Phase 4: Proposed Architecture and Review 1 (October)**

- **Week 1-2:** Draw the proposed architecture using draw.io tool and mention the modules in a detailed way.

- **Week 3-4:** Review 1 submission that has the finalized problem statement, objectives, scope, and system architecture.

**Phase 5: Implementation (November)**

- **Week 1-2:** Install the dependencies in the Visual Studio code. Create 2 folders- frontend and backend.
- **Week 3-4:** Start working with the frontend code using React.js and create home, contact, about, booking pages, etc.

**Phase 6: Backend Implementation (December)**

- **Week 1-2:** Complete 70% of implementation part and check the website using the command "npm run dev".
- **Week 3-4:** Initiate the backend using Express.js and add a Dotenv file for creating environment variables, such as database connection strings, API keys, port number and secret tokens.

**Phase 7: Review 2 (January)**

- **Week 1-2:** Prepare for Review 2 components and complete the implementation to check the further improvements.
- **Week 3-4:** Start the Review 2 documentation according to the schedule.

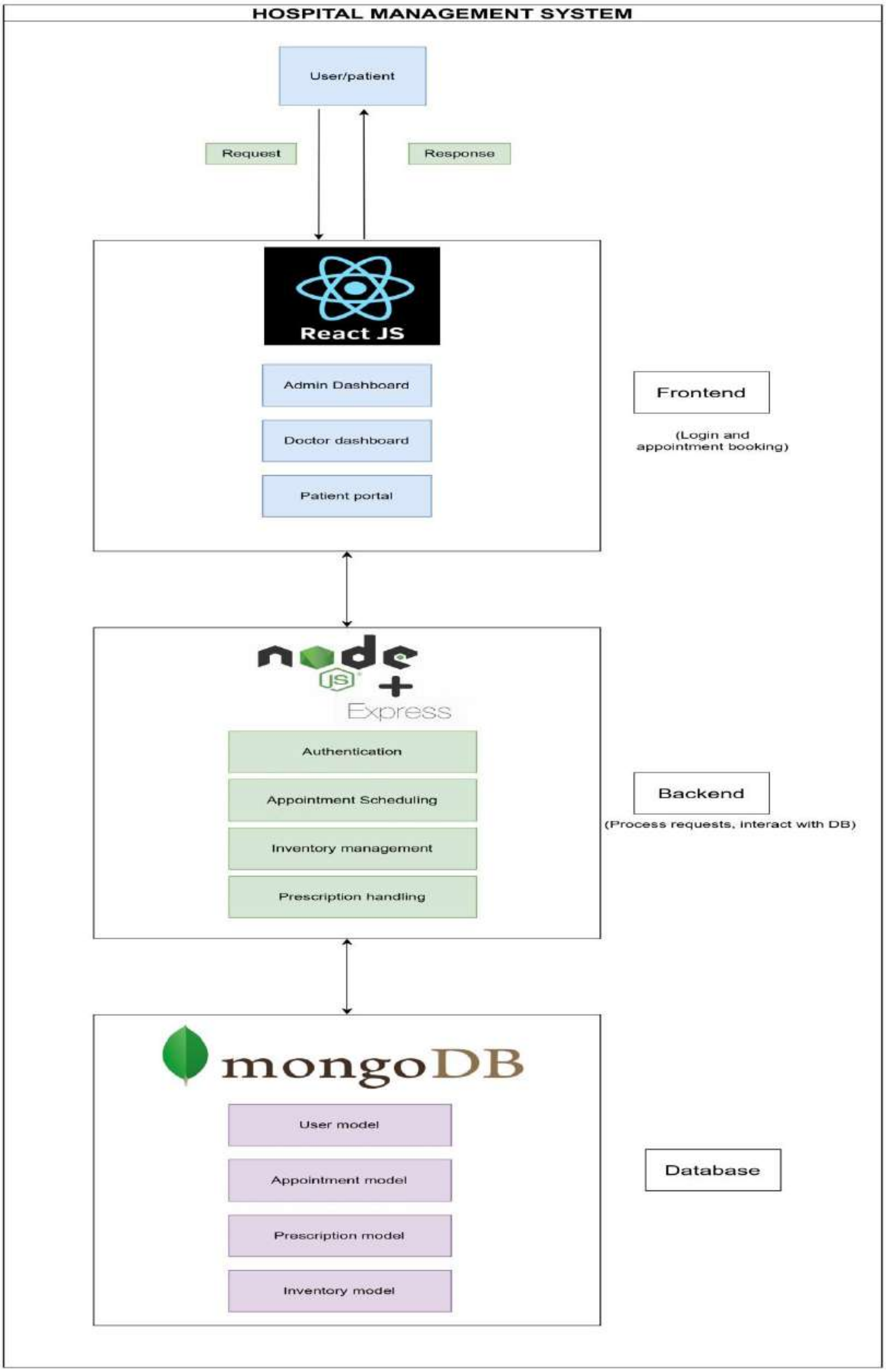
**Final Phase: Review 3 (February)**

- **Week 1-2:** Complete the project documentation with other contents and presenting to project guide to incorporate if any changes required.
- **Week 3-4:** Final Review presentation to the allotted panel.

The Literature Survey papers are added to my Github page repositories. The roadmap of the project and issues related to the project are created in the project.

# 3. SYSTEM DESIGN

## 3.1 PROPOSED ARCHITECTURE



## 3.2 MODULE DESCRIPTION

### 3.2.1 User

User initiates login/registration and creates an account in the website. The request to book an appointment is sent to the frontend and frontend works with backend and database to collect the required data to send appointment availability and booking details to the patient.

### 3.2.2 Frontend

**React JS:** Serves as the patient and hospital staff's user interface (UI). It shows data from the MongoDB database and communicates with the backend through APIs.

**Components:**

- **Admin Dashboard:** Provides access to hospital data, medicines, and doctor appointment scheduling management for administrators.
- **Doctor Dashboard:** This allows doctors to manage prescriptions, see appointments, and work with patient information.
- **Patient portal:** Prescriptions, medical history, and appointment scheduling are all available to patients.

### 3.2.3 Backend

**Node JS and Express JS:** Manages all server-side logic, responds to front-end API calls, and interacts with the MongoDB database.

**Components:**

- **Authentication:** Uses JWT (JSON Web Tokens) to administer and verify access control for the roles of Admin, Doctor, and Patient.
- **Appointment Scheduling:** Takes care of making reservations, organizing doctor calendars, and informing patients of availability.
- **Inventory management:** Admin have the ability to track, add, and edit medications.
- **Prescription handling:** Doctors are able to write and amend prescriptions for their patients.

### 3.2.4 Database

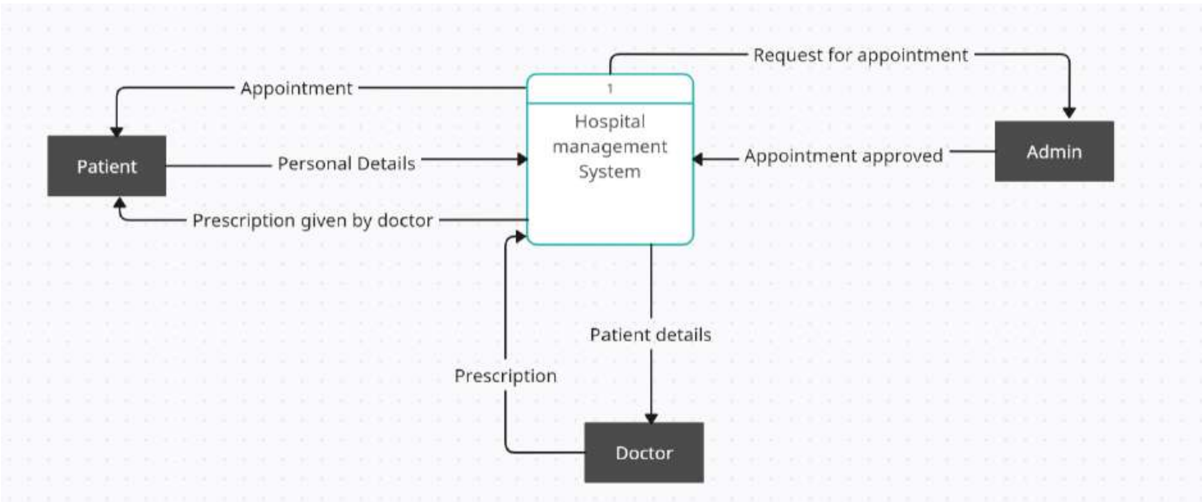
**Mongo DB:** Keeps track of every application's data, such as patient information, appointment and doctor schedules, prescription information, and inventory details.

**Components:**

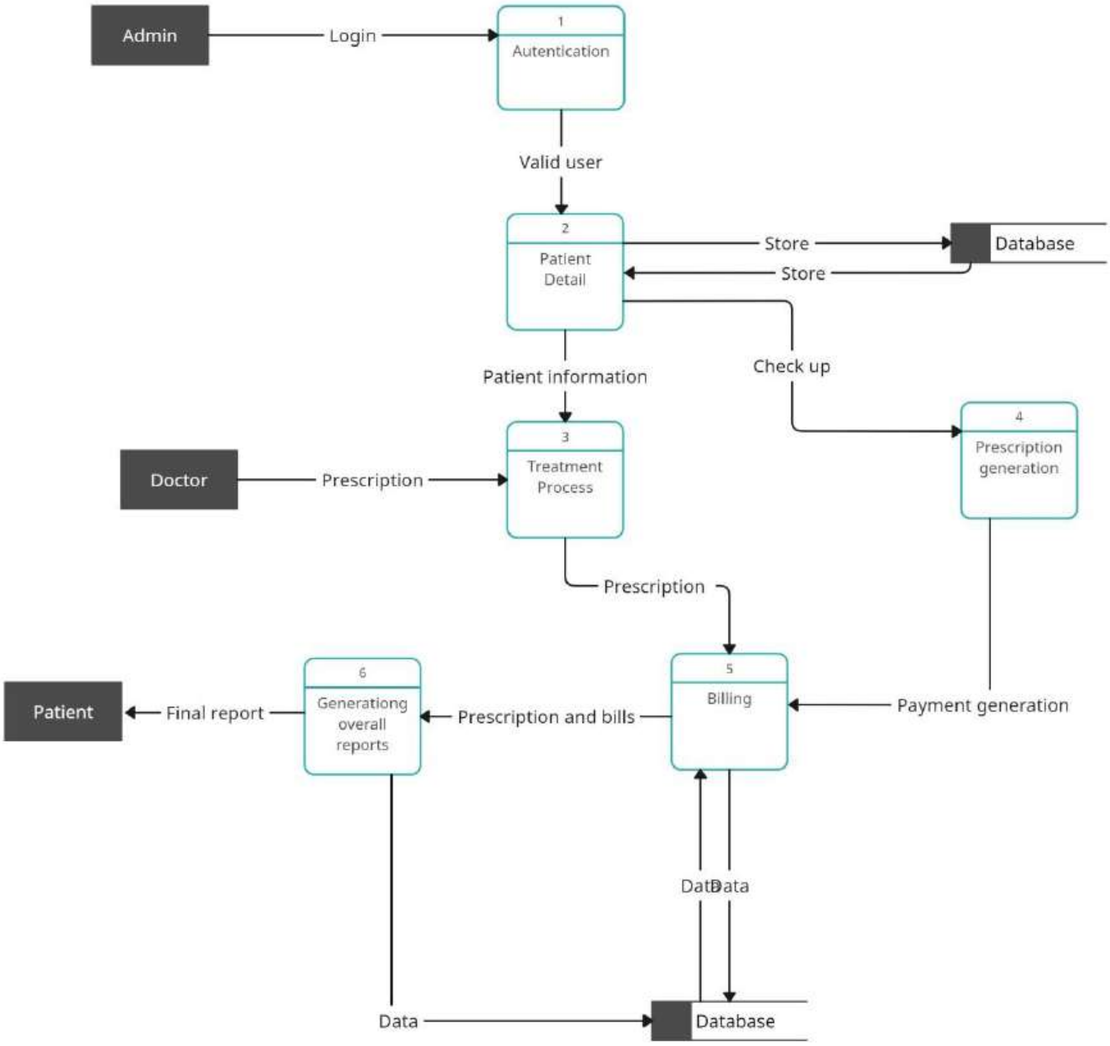
- **User Model:** Provides details (name, contact information, role, etc.) about Admins, Doctors, and Patients.
- **Appointment Model:** Keeps track of patient, doctor, date, and status information for each appointment.
- **Prescription Model:** Stores any medications that doctors prescribe for patients as well as their medical prescriptions.
- **Inventory Model:** Maintains track of available medications, stock amounts, and status of reorders.

### 3.3 DFD FOR EACH MODULE

**DFD Level 0:**

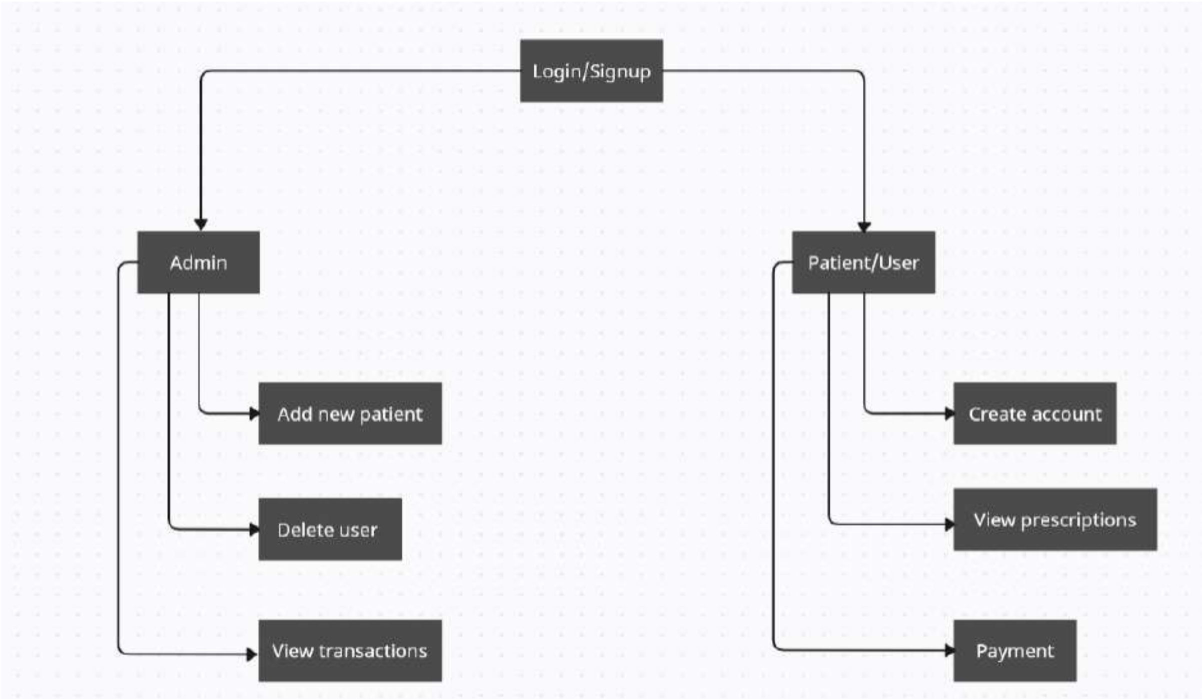


**DFD Level 1:**

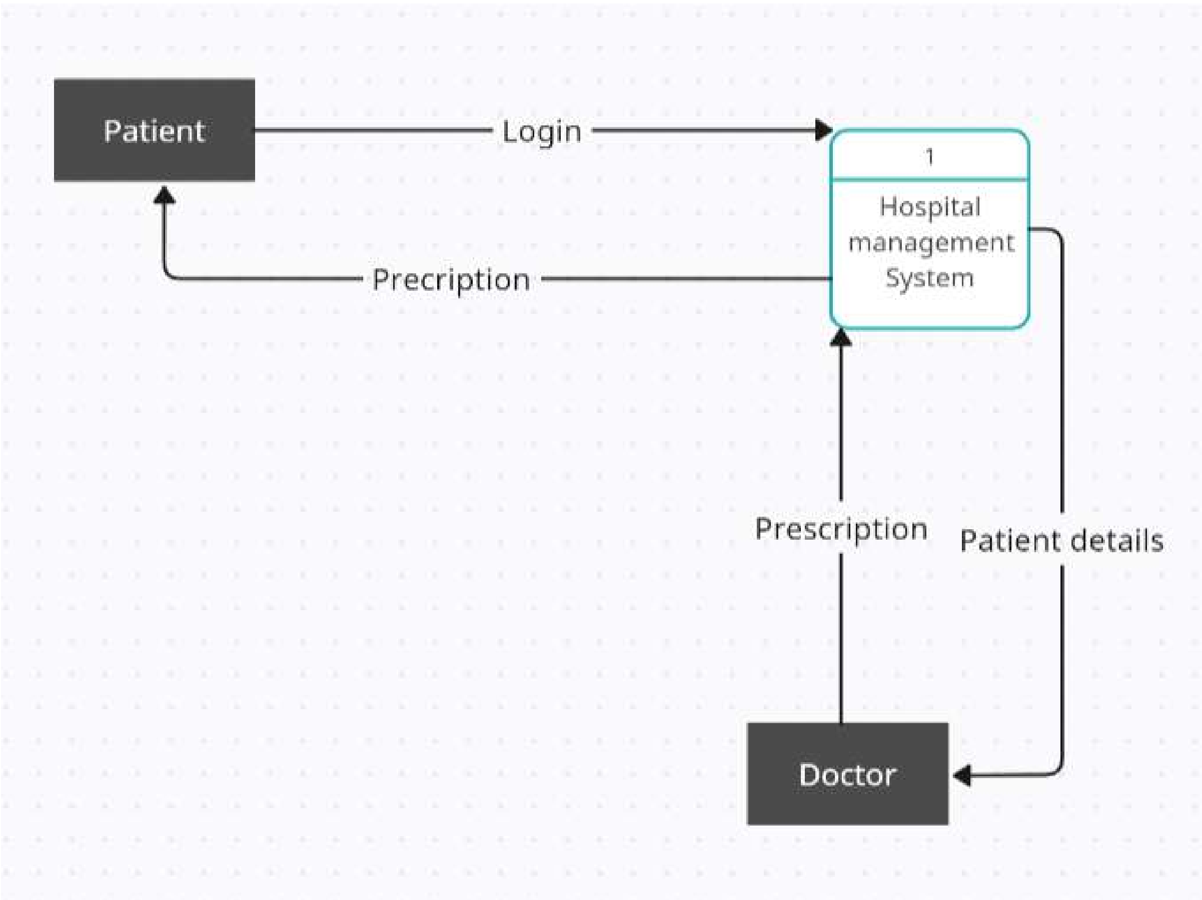


Admin
Login
1
Autentication
Valid user
2
Patient Detail
Store
Database
Store
Check up
Patient information
3
Treatment Process
Doctor
Prescription
4
Prescription generation
Prescription
5
Billing
Payment generation
6
Generationg overall reports
Prescription and bills
Patient
Final report
DatData
Data
Database

**DFD Level 2 for login:**




**DFD Level 2 for doctor giving prescription:**



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